WIKIPEDIA 25
25 years of the free encyclopedia

# List of file signatures

A file signature is data used to identify or verify the content of a file. Such signatures are also known as *magic numbers* or *magic bytes* and are usually inserted at the beginning of the file.

Many file formats are not intended to be read as text. If such a file is accidentally viewed as a text file, its contents will be unintelligible. However, some file signatures can be recognizable when interpreted as text. In the table below, the column "ISO 8859-1" shows how the file signature appears when interpreted as text in the common ISO 8859-1 encoding, with unprintable characters represented as the control code abbreviation or symbol, or codepage 1252 character where available, or a box otherwise. In some cases the space character is shown as SP.

| Hex signature | ISO 8859-1 | Offset | Extension | Description |
|---|---|---|---|---|
| 23 21 | #! | 0 | | Script or data to be passed to the program following the shebang (#!)[1] |
| 02 00 5a 57 52 54 00 00 00 00 00 00 00 00 00 00 | ZWRT | 0 | cwk | Claris Works word processing doc |
| 00 00 02 00 06 04 06 00 08 00 00 00 00 00 | | 0 | wk1 | Lotus 1-2-3 spreadsheet (v1) file |
| 00 00 1A 00 00 10 04 00 00 00 00 00 | | 0 | wk3 | Lotus 1-2-3 spreadsheet (v3) file |
| 00 00 1A 00 02 10 04 00 00 00 00 00 | | 0 | wk4<br>wk5 | Lotus 1-2-3 spreadsheet (v4, v5) file |
| 00 00 1A 00 05 10 04 | | 0 | 123 | Lotus 1-2-3 spreadsheet (v9) file |
| 00 00 03 F3 | ó | 0 | | Amiga Hunk executable file |
| 00 00 49 49 58 50 52 (little-endian)<br>00 00 4D 4D 58 50 52 (big-endian) | IIXPR<br>MMXPR | 0 | qxd | Quark Express document |
| 50 57 53 33 | PWS3 | 0 | psafe3 | Password Gorilla Password Database |
| D4 C3 B2 A1 (little-endian) | ÔÃ²¡ | 0 | pcap | Libpcap File Format[2] |
| A1 B2 C3 D4 (big-endian) | ¡²ÃÔ | | | |
| 4D 3C B2 A1 (little-endian) | M<²¡ | 0 | pcap | Libpcap File Format (nanosecond-resolution) [2] |
| A1 B2 3C 4D (big-endian) | ¡²<M | | | |
| 0A 0D 0D 0A | | 0 | pcapng | PCAP Next Generation Dump File Format[3] |
| ED AB EE DB | í«îÛ | 0 | rpm | RedHat Package Manager (RPM) package[4] |
| 53 51 4C 69 74 65 20 66 6F 72 6D 61 74 20 33 00 | SQLite format 3 | 0 | sqlitedb<br>sqlite<br>db | SQLite Database[5] |

| Hex signature | ISO 8859-1 | Offset | Extension | Description |
|---|---|---|---|---|
| 53 50 30 31 | SP01 | 0 | bin | Amazon Kindle Update Package[6] |
| 49 57 41 44 | IWAD | 0 | wad | internal WAD (main resource file of Doom)[7] |
| 00 | ␀ | 0 | PIC<br>PIF<br>SEA<br>YTR | IBM Storyboard bitmap file<br>Windows Program Information File<br>Mac Stuffit Self-Extracting Archive<br>IRIS OCR data file |
| 00 00 00 00 00 00 00<br>00 00 00 00 00 00 00<br>00 00 00 00 00 00 00 | ␀␀␀␀␀␀␀␀<br>␀␀␀␀␀␀␀␀<br>␀␀␀␀␀␀␀␀ | 11 | PDB | PalmPilot Database/ Document File |
| BE BA FE CA | ¾ºþÊ | 0 | DBA | Palm Desktop Calendar Archive |
| 00 01 42 44 | ␀␁BD | 0 | DBA | Palm Desktop To Do Archive |
| 00 01 44 54 | ␀␁DT | 0 | TDA | Palm Desktop Calendar Archive |
| 54 44 46 24 | TDF$ | 0 | TDF$ | Telegram Desktop File |
| 54 44 45 46 | TDEF | 0 | TDEF | Telegram Desktop Encrypted File |
| 00 01 00 00 | ␀␁␀␀ | 0 | | Palm Desktop Data File (Access format) |
| 00 00 01 00 | ␀␀␁␀ | 0 | ico | Computer icon encoded in ICO file format[8] |
| 69 63 6e 73 | icns | 0 | icns | Apple Icon Image format |
| 66 74 79 70 33 67 | ftyp3g | 4 | 3gp<br>3g2 | 3rd Generation Partnership Project 3GPP and 3GPP2 multimedia files |
| 66 74 79 70 68 65 69 6366 74 79 70 6d | ftypheic | 4 | heic | High Efficiency Image Container (HEIC) |
| 1F 9D | ␟□ | 0 | z<br>tar.z | compressed file (often tar zip) using Lempel-Ziv-Welch algorithm |
| 1F A0 | ␟␠ | 0 | z<br>tar.z | Compressed file (often tar zip) using LZH algorithm |

| Hex signature | ISO 8859-1 | Offset | Extension | Description |
|---|---|---|---|---|
| 2D 6C 68 30 2D | -lh0- | 2 | lzh | Lempel Ziv Huffman archive file Method 0 (No compression) |
| 2D 6C 68 35 2D | -lh5- | 2 | lzh | Lempel Ziv Huffman archive file Method 5 (8 KiB sliding window) |
| 42 41 43 4B 4D 49 4B 45 44 49 53 4B | BACKMIKEDISK | 0 | bac | AmiBack Amiga Backup data file |
| 49 4E 44 58 | INDX | 0 | idx | AmiBack Amiga Backup index file |
| 62 70 6C 69 73 74 | bplist | 0 | plist | Binary Property List file |
| 42 5A 68 | BZh | 0 | bz2 | Compressed file using Bzip2 algorithm |
| 47 49 46 38 37 61<br>47 49 46 38 39 61 | GIF87a<br>GIF89a | 0 | gif | Image file encoded in the Graphics Interchange Format (GIF)[9] |
| 49 49 2A 00 (little-endian) | II*␀ | 0 | tif<br>tiff | Tagged Image File Format (TIFF)[10] |
| 4D 4D 00 2A (big-endian) | MM␀* | | | |
| 49 49 2B 00 (little-endian) | II+␀ | 0 | tif<br>tiff | BigTIFF[11] |
| 4D 4D 00 2B (big-endian) | MM␀+ | 0 | | |
| 49 49 2A 00 10 00 00 00 43 52 | II*␀␐␀␀␀CR | 0 | cr2 | Canon RAW Format Version 2[12]<br>Canon's RAW format is based on TIFF.[13] |
| 66 74 79 70 63 72 78 43 52 | ftypcrx | 0 | cr3 | |
| 80 2A 5F D7 | €*_× | 0 | cin | Kodak Cineon image |
| 52 4E 43 01<br>52 4E 43 02 | RNC␁<br>RNC␂ | 0 | | Compressed file using Rob Northen Compression (version 1 and 2) algorithm[14] |
| 4E 55 52 55 49 4D 47<br>4E 55 52 55 50 41 4C | NURUIMG<br>NURUPAL | 0 | nui<br>nup | nuru ASCII/ANSI image and palette files[15] |
| 53 44 50 58 (big-endian | SDPX | 0 | dpx | SMPTE DPX image |

| Hex signature | ISO 8859-1 | Offset | Extension | Description |
|---|---|---|---|---|
| format) | | | | |
| 58 50 44 53 (little-endian format) | XPDS | | | |
| 76 2F 31 01 | v/1 | 0 | exr | OpenEXR image |
| 42 50 47 FB | BPGû | 0 | bpg | Better Portable Graphics format[16] |
| FF D8 FF DB | ÿØÿÛ | 0 | jpg<br>jpeg | JPEG raw or in the JFIF or Exif file format[17] |
| FF D8 FF E0 00 10 4A 46 49 46 00 01 | ÿØÿà JFIF | | | |
| FF D8 FF EE | ÿØÿî | | | |
| FF D8 FF E1 ?? ?? 45 78 69 66 00 00 | ÿØÿá??Exif | | | |
| FF D8 FF E0 | ÿØÿà | 0 | jpg | JPEG raw or in the JFIF or Exif file format[17] |
| 00 00 00 0C 6A 50 20 20 0D 0A 87 0A | jP | 0 | jp2<br>j2k<br>jpf<br>jpm<br>jpg2<br>j2c<br>jpc<br>jpx<br>mj2 | JPEG 2000 format[18] |
| FF 4F FF 51 | ÿOÿQ | | | |
| 71 6F 69 66 | qoif | 0 | qoi | QOI - The “Quite OK Image Format”[19] |
| 46 4F 52 4D ?? ?? ?? ?? 49 4C 42 4D | FORM????ILBM | 0<br>any | ilbm<br>lbm<br>ibm<br>iff | IFF Interleaved Bitmap Image |
| 46 4F 52 4D ?? ?? ?? ?? 38 53 56 58 | FORM????8SVX | 0<br>any | 8svx<br>8sv<br>svx<br>snd<br>iff | IFF 8-Bit Sampled Voice |
| 46 4F 52 4D ?? ?? ?? ?? 41 43 42 4D | FORM????ACBM | 0<br>any | acbm<br>iff | Amiga Contiguous Bitmap |
| 46 4F 52 4D ?? ?? ?? ?? 41 4E 42 4D | FORM????ANBM | 0<br>any | anbm<br>iff | IFF Animated Bitmap |
| 46 4F 52 4D ?? ?? ?? ?? 41 4E 49 4D | FORM????ANIM | 0<br>any | anim<br>iff | IFF CEL Animation |

| Hex signature | ISO 8859-1 | Offset | Extension | Description |
|---|---|---|---|---|
| 46 4F 52<br>4D ?? ?? ?? ??<br>46 41 58 58 | FORM????FAXX | 0<br>any | faxx<br>fax<br>iff | IFF Facsimile Image |
| 46 4F 52<br>4D ?? ?? ?? ??<br>46 54 58 54 | FORM????FTXT | 0<br>any | ftxt<br>iff | IFF Formatted Text |
| 46 4F 52<br>4D ?? ?? ?? ??<br>53 4D 55 53 | FORM????SMUS | 0<br>any | smus<br>smu<br>mus<br>iff | IFF Simple Musical Score |
| 46 4F 52<br>4D ?? ?? ?? ??<br>43 4D 55 53 | FORM????CMUS | 0<br>any | cmus<br>mus<br>iff | IFF Musical Score |
| 46 4F 52<br>4D ?? ?? ?? ??<br>59 55 56 4E | FORM????YUVN | 0<br>any | yuvn<br>yuv<br>iff | IFF YUV Image |
| 46 4F 52<br>4D ?? ?? ?? ??<br>46 41 4E 54 | FORM????FANT | 0<br>any | iff | Amiga Fantavision Movie |
| 46 4F 52<br>4D ?? ?? ?? ??<br>41 49 46 46 | FORM????AIFF | 0<br>any | aiff<br>aif<br>aifc<br>snd<br>iff | Audio Interchange File Format |
| 4C 5A 49 50 | LZIP | 0 | lz | lzip compressed file[20] |
| 30 37 30 37 30 31 | 070701 | 0 | cpio | cpio "new" ASCII archive file |
| 30 37 30 37 30 32 | 070702 | 0 | cpio | cpio ASCII archive file with crc |
| 30 37 30 37 30 37 | 070707 | 0 | cpio | cpio ASCII archive file[21] |
| 4D 5A | MZ | 0 | exe<br>dll<br>mui<br>sys<br>scr<br>cpl<br>ocx<br>ax<br>iec<br>ime<br>rs<br>tsp<br>fon<br>efi | DOS MZ executable and its descendants (including NE and PE) |
| 53 4D 53 4E 46 32 30 30 | SMSNF200 | 0 | ssp | SmartSniff Packets File [22] |

| Hex signature | ISO 8859-1 | Offset | Extension | Description |
|---|---|---|---|---|
| 5A 4D | ZM | 0 | exe | DOS ZM executable and its descendants (rare) |
| 50 4B 03 04<br>50 4B 05 06 (empty archive)<br>50 4B 07 08 (spanned archive) | PK<sup>ETX EOT</sup><br>PK<sup>ENQ ACK</sup><br>PK<sup>BEL BS</sup> | 0 | zip<br>aar<br>apk<br>docx<br>epub<br>ipa<br>jar<br>kmz<br>maff<br>msix<br>odp<br>ods<br>odt<br>pk3<br>pk4<br>pptx<br>usdz<br>vsdx<br>xlsx<br>xpi<br>whl | zip file format and formats based on it, such as EPUB, JAR, ODF, OOXML |
| 52 61 72 21 1A 07 00 | Rar!<sup>SUB BEL NUL</sup> | 0 | rar | Roshal ARchive compressed archive v1.50 onwards[23] |
| 52 61 72 21 1A 07 01 00 | Rar!<sup>SUB BEL SOH NUL</sup> | 0 | rar | Roshal ARchive compressed archive v5.00 onwards[24] |
| 7F 45 4C 46 | <sup>DEL</sup>ELF | 0 | none, .axf, .bin, .elf, .o, .out, .prx, .puff, .ko, .mod, .so | Executable and Linkable Format[25] |
| 89 50 4E 47 0D 0A 1A 0A | %PNG<sup>CR LF SUB LF</sup> | 0 | png | Image encoded in the Portable Network Graphics format[26] |
| 0E 03 13 01 | <sup>ENQ ETX DC3 SOH</sup> | 0 | hdf4<br>h4 | Data stored in version 4 of the Hierarchical Data Format. |
| 89 48 44 46 0D 0A 1A 0A | %HDF<sup>CR LF SUB LF</sup> | 0, 512, 1024, 2048, ... | hdf5<br>h5 | Data stored in version 5 of the Hierarchical Data Format. |
| C9 | É | 0 | com | CP/M 3 and higher with overlays[27] |
| CA FE BA BE | Êþº¾ | 0 | class | Java class file, Mach-O Fat Binary |
| EF BB BF | ï»¿ | 0 | txt<br>*others* | UTF-8 byte order mark, commonly seen in text files.[28][29][30] |

| Hex signature | ISO 8859-1 | Offset | Extension | Description |
|---|---|---|---|---|
| FF FE | ÿþ | 0 | txt<br>*others* | UTF-16LE byte order mark, commonly seen in text files.[28][29][30] |
| FE FF | þÿ | 0 | txt<br>*others* | UTF-16BE byte order mark, commonly seen in text files.[28][29][30] |
| FF FE 00 00 | ÿþ¼¼ | 0 | txt<br>*others* | UTF-32LE byte order mark for text[28][30] |
| 00 00 FE FF | ¼¼þÿ | 0 | txt<br>*others* | UTF-32BE byte order mark for text[28][30] |
| 2B 2F 76 38<br>2B 2F 76 39<br>2B 2F 76 2B<br>2B 2F 76 2F | +/v8<br>+/v9<br>+/v+<br>+/v/ | 0 | | UTF-7 byte order mark for text[31][30] |
| 0E FE FF | ␎þÿ | 0 | txt<br>*others* | SCSU byte order mark for text[31][30] |
| DD 73 66 73 | Ýsfs | 0 | | UTF-EBCDIC byte order mark for text[31][30] |
| FE ED FA CE | þíúÎ | 0<br>0x1000 | | Mach-O binary (32-bit) |
| FE ED FA CF | þíúÏ | 0<br>0x1000 | | Mach-O binary (64-bit) |
| FE ED FE ED | þíþí | 0 | | JKS Javakey Store[32] |
| CE FA ED FE | Îúíþ | 0 | | Mach-O binary (reverse byte ordering scheme, 32-bit)[33] |
| CF FA ED FE | Ïúíþ | 0 | | Mach-O binary (reverse byte ordering scheme, 64-bit)[33] |
| 25 21 50 53 | %!PS | 0 | ps | PostScript document |
| 25 21 50 53 2D 41 64 6F 62 65 2D 33 2E 30 20 45 50 53 46 2D 33 2E 30 | %!PS-Adobe-3.0 ESPF-3.0 | 0 | eps<br>epsf | Encapsulated PostScript file version 3.0[34] |
| 25 21 50 53 2D 41 64 6F 62 65 2D 33 2E 31 20 45 50 53 46 2D 33 2E 30 | %!PS-Adobe-3.1 ESPF-3.0 | 0 | eps<br>epsf | Encapsulated PostScript file version 3.1[35] |
| 49 54 53 46 03 00 00 00 60 00 00 00 | ITSF¾¼¼¼`¼¼¼ | 0 | chm | MS Windows HtmlHelp Data |
| 3F 5F | ?_ | 0 | hlp | Windows 3.x/95/98 Help file |

| Hex signature | ISO 8859-1 | Offset | Extension | Description |
|---|---|---|---|---|
| 25 50 44 46 2D | %PDF- | 0 | pdf | PDF document[36] |
| 30 26 B2 75 8E 66 CF 11 A6 D9 00 AA 00 62 CE 6C | 0&²uŽfÏ¦Ù ª bÎl | 0 | asf<br>wma<br>wmv | Advanced Systems Format[37] |
| 24 53 44 49 30 30 30 31 | $SDI0001 | 0 | | System Deployment Image, a disk image format used by Microsoft |
| 4F 67 67 53 | OggS | 0 | ogg<br>oga<br>ogv | Ogg, an open source media container format |
| 38 42 50 53 | 8BPS | 0 | psd | Photoshop Document file, Adobe Photoshop's native file format |
| 52 49 46 46 ?? ?? ?? ?? 57 41 56 45 | RIFF????WAVE | 0 | wav | Waveform Audio File Format[38] |
| 52 49 46 46 ?? ?? ?? ?? 41 56 49 20 | RIFF????AVI ₛₚ | 0 | avi | Audio Video Interleave video format[39] |
| FF FB<br>FF F3<br>FF F2 | ÿû<br>ÿó<br>ÿò | 0 | mp3 | MPEG-1 Layer 3 file without an ID3 tag or with an ID3v1 tag (which is appended at the end of the file) |
| 49 44 33 | ID3 | 0 | mp3 | MP3 file with an ID3v2 container |
| 42 4D | BM | 0 | bmp<br>dib | BMP file, a bitmap format used mostly in the Windows world |
| 43 44 30 30 31 | CD001 | 0x8001<br>0x8801<br>0x9001 | iso | ISO9660 CD/DVD image file[40] |
| 43 44 30 30 31 | CD001 | 0x5EAC9 | cdi | CD-i CD image file |
| 6D 61 69 6E 2E 62 73 | main.bs | 0 | mgw | Nintendo Game & Watch image file |
| 4E 45 53 | NES | 0 | nes | Nintendo Entertainment System image file |
| A0 32 41 A0 A0 A0 | 2A | 0x165A4 | d64 | Commodore 64 1541 disk image (D64 format) |
| 47 53 52 2D 31 35 34 31 | GCR-1541 | 0 | g64 | Commodore 64 1541 disk image (G64 format) |
| A0 33 44 A0 A0 | 3D | 0x61819 | d81 | Commodore 64 1581 disk image (D81 format) |

| Hex signature | ISO 8859-1 | Offset | Extension | Description |
|---|---|---|---|---|
| 43 36 34 20 74 61 70 65 20 69 6D 61 67 65 20 66 69 6C 65 | C64 tape image file | 0 | t64 | Commodore 64 tape image |
| 43 36 34 20 43 41 52 54 52 49 44 47 45 20 20 20 | C64 CARTRIDGE   | 0 | crt | Commodore 64 cartridge image |
| 53 49 4D 50 4C 45 20 20<br>3D 20 20 20 20 20 20 20<br>20 20 20 20 20 20 20 20<br>20 20 20<br>20 20 20 20 20 54 | SIMPLE  <br>=       <br>          <br>T | 0 | fits | Flexible Image Transport System (FITS)[41] |
| 66 4C 61 43 | fLaC | 0 | flac | Free Lossless Audio Codec[42] |
| 4D 54 68 64 | MThd | 0 | mid<br>midi | MIDI sound file[43] |
| D0 CF 11 E0 A1 B1 1A E1 | ÐÏ à¡±á | 0 | doc<br>xls<br>ppt<br>msi<br>msg | Compound File Binary Format, a container format defined by Microsoft COM. It can contain the equivalent of files and directories. It is used by Windows Installer and for documents in older versions of Microsoft Office.[44] It can be used by other programs as well that rely on the COM and OLE API's. (0xD0CF11E0 visually resembles "DOCFILE".[45]) |
| 64 65 78 0A 30 33 35 00 | dex 035 | 0 | dex | Dalvik Executable |
| 4B 44 4D | KDM | 0 | vmdk | VMDK files[46][47] |
| 23 20 44 69 73 6B 20 44 65 73 63 72 69 70 74 6F | # Disk Descripto | 0 | vmdk | VMware 4 Virtual Disk description file (split disk) |
| 43 72 32 34 | Cr24 | 0 | crx | Google Chrome extension[48] or packaged app[49] |
| 41 47 44 33 | AGD3 | 0 | fh8 | FreeHand 8 document[50][51] |

| Hex signature | ISO 8859-1 | Offset | Extension | Description |
|---|---|---|---|---|
| 05 07 00 00 42 4F 42 4F<br>05 07 00 00 00 00 00 00<br>00 00 00 00 00 01 | BOBO | 0 | cwk | AppleWorks 5 document |
| 06 07 E1 00 42 4F 42 4F<br>06 07 E1 00 00 00 00 00<br>00 00 00 00 00 01 | áBOBO | 0 | cwk | AppleWorks 6 document |
| 45 52 02 00 00 00 | ER | 0 | toast | Roxio Toast disc image file |
| 8B 45 52 02 00 00 00 | ‹ER | | | |
| 6B 6F 6C 79 | koly | end–512 | dmg | Apple Disk Image file |
| 78 61 72 21 | xar! | 0 | xar | eXtensible ARchive format[52] |
| 50 4D 4F 43 43 4D 4F 43 | PMOCCMOC | 0 | dat | Windows Files And Settings Transfer Repository[53] See also USMT 3.0 (Win XP)[54] and USMT 4.0 (Win 7)[55] User Guides |
| 4E 45 53 1A | NES | 0 | nes | Nintendo Entertainment System ROM file[56] |
| 75 73 74 61 72 00 30 30<br>75 73 74 61 72 20 20 00 | ustar00<br>ustar | 257 | tar | tar archive[57] |
| 4F 41 52 ?? | OAR? | 0 | oar | OAR file archive format, where ?? is the format version. |
| 74 6F 78 33 | tox3 | 0 | tox | Open source portable voxel file[58] |
| 4D 4C 56 49 | MLVI | 0 | MLV | Magic Lantern Video file[59] |
| 44 43 4D 01 50 41 33 30<br>50 41 33 30 | DCMPA30<br>PA30 | 0 | | Windows Update Binary Delta Compression file[60] |
| 37 7A BC AF 27 1C | 7z¼¯'' | 0 | 7z | 7-Zip File Format |
| 1F 8B | ‹ | 0 | gz<br>tar.gz | GZIP compressed file[61] |

| Hex signature | ISO 8859-1 | Offset | Extension | Description |
|---|---|---|---|---|
| FD 37 7A 58 5A 00 | ý7zXZ | 0 | xz<br>tar.xz | XZ compression utility using LZMA2 compression |
| 04 22 4D 18 | "M | 0 | lz4 | LZ4 Frame Format[62]<br>Remark: LZ4 block format does not offer any magic bytes.[63] |
| 4D 53 43 46 | MSCF | 0 | cab | Microsoft Cabinet file |
| 53 5A 44 44 88 F0 27 33 | SZDD^ð'3 | 0 | ??_ | Microsoft compressed file in Quantum format, used prior to Windows XP. File can be decompressed using Extract.exe or Expand.exe distributed with earlier versions of Windows. After compression, the last character of the original filename extension is replaced with an underscore, e.g. 'Setup.exe' becomes 'Setup.ex_'. |
| 46 4C 49 46 | FLIF | 0 | flif | Free Lossless Image Format |
| 1A 45 DF A3 | Eß£ | 0 | mkv<br>mka<br>mks<br>mk3d<br>webm | Matroska media container, including WebM |
| 4D 49 4C 20 | MIL | 0 | stg | "SEAN : Session Analysis" Training file. Also used in compatible software "Rpw : Rowperfect for Windows" and "RP3W : ROWPERFECT3 for Windows". |
| 41 54 26 54 46 4F 52 4D ?? ?? ?? ?? 44 4A 56 | AT&TFORM????DJV | 0 | djvu<br>djv | DjVu document<br>The following byte is either 55 (U) for single-page or 4D (M) for multi-page documents. |
| 2D 2D 2D 2D 2D 42 45 47 49 4E 20 43 45 52 54 49 46 49 43 41 54 45 2D 2D 2D 2D 2D | -----BEGIN CERTIFICATE----- | 0 | crt<br>pem | PEM encoded X.509 certificate |

| Hex signature | ISO 8859-1 | Offset | Extension | Description |
|---|---|---|---|---|
| 2D 2D 2D 2D 2D 42 45 47 49 4E 20 43 45 52 54 49 46 49 43 41 54 45 20 52 45 51 55 45 53 54 2D 2D 2D 2D 2D | -----BEGIN CERTIFICATE REQUEST----- | 0 | csr<br>pem | PEM encoded X.509 Certificate Signing Request |
| 2D 2D 2D 2D 2D 42 45 47 49 4E 20 50 52 49 56 41 54 45 20 4B 45 59 2D 2D 2D 2D 2D | -----BEGIN PRIVATE KEY----- | 0 | key<br>pem | PEM encoded X.509 PKCS#8 private key |
| 2D 2D 2D 2D 2D 42 45 47 49 4E 20 44 53 41 20 50 52 49 56 41 54 45 20 4B 45 59 2D 2D 2D 2D 2D | -----BEGIN DSA PRIVATE KEY----- | 0 | key<br>pem | PEM encoded X.509 PKCS#1 DSA private key |
| 2D 2D 2D 2D 2D 42 45 47 49 4E 20 52 53 41 20 50 52 49 56 41 54 45 20 4B 45 59 2D 2D 2D 2D 2D | -----BEGIN RSA PRIVATE KEY----- | 0 | key<br>pem | PEM encoded X.509 PKCS#1 RSA private key |
| 50 75 54 54 59 2D 55 73 65 72 2D 4B 65 79 2D 46 69 6C 65 2D 32 3A | PuTTY-User-Key-File-2: | 0 | ppk | PuTTY private key file version 2 |
| 50 75 54 54 59 2D 55 73 65 72 2D 4B 65 79 2D 46 69 6C 65 2D 33 3A | PuTTY-User-Key-File-3: | 0 | ppk | PuTTY private key file version 3 |
| 2D 2D 2D 2D 2D 42 45 47 49 4E 20 4F 50 45 4E 53 53 48 20 50 52 49 56 41 54 45 20 4B 45 59 2D 2D 2D 2D 2D | -----BEGIN OPENSSH PRIVATE KEY----- | 0 | | OpenSSH private key file |
| 2D 2D 2D 2D 2D 42 45 47 49 4E 20 53 53 48 32 20 4B 45 59 2D 2D 2D 2D 2D | -----BEGIN SSH2 PUBLIC KEY----- | 0 | pub | OpenSSH public key file |

| Hex signature | ISO 8859-1 | Offset | Extension | Description |
|---|---|---|---|---|
| 44 49 43 4D | DICM | 128 | dcm | DICOM Medical File Format |
| 77 4F 46 46 | wOFF | 0 | woff | WOFF File Format 1.0[64] |
| 77 4F 46 32 | wOF2 | 0 | woff2 | WOFF File Format 2.0[65] |
| 3C 3F 78 6D 6C 20 | <?xml␣ | 0 after BOM | xml (UTF-8 or other 8-bit encodings) | eXtensible Markup Language[29][66] |
| 3C 00 3F 00 78 00 6D 00 6C 00 20 | <␀?␀x␀m␀l␀␣␀ | | xml (UTF-16LE) | |
| 00 3C 00 3F 00 78 00 6D 00 6C 00 20 | ␀<␀?␀x␀m␀l␀␣ | | xml (UTF-16BE) | |
| 3C 00 00 00 3F 00 00 00 78 00 00 00 6D 00 00 00 6C 00 00 00 20 00 00 00 | <␀␀␀?␀␀␀ x␀␀␀m␀␀␀ l␀␀␀␣␀␀␀ | | xml (UTF-32LE) | |
| 00 00 00 3C 00 00 00 3F 00 00 00 78 00 00 00 6D 00 00 00 6C 00 00 00 20 | ␀␀␀<␀␀␀? ␀␀␀x␀␀␀m ␀␀␀l␀␀␀␣ | | xml (UTF-32BE) | |
| 4C 6F A7 94 93 40 | Lo§”“@ | | xml (EBCDIC) | |
| 00 61 73 6D | ␀asm | 0 | wasm | WebAssembly binary format[67] |
| CF 84 01 | Ï„␁ | 0 | lep | Lepton compressed JPEG image[68] |
| 43 57 53 | CWS | 0 | swf | Adobe Flash .swf |
| 46 57 53 | FWS | | | |
| 21 3C 61 72 63 68 3E 0A | !<arch>␤ | 0 | deb | linux deb file |
| C0 3B 39 98 | À;9˜ | 0 | | Linux EXT3/EXT4 jbd2 filesystem Journal[69] |
| 52 49 46 46 ?? ?? ?? ?? 57 45 42 50 | RIFF????WEBP | 0 | webp | Google WebP image file, where ?? ?? ?? ?? is the file size. More information on WebP File Header (https://developers.google.com/speed/webp/docs/riff_container#webp_file_header) |

| Hex signature | ISO 8859-1 | Offset | Extension | Description |
|---|---|---|---|---|
| 27 05 19 56 | 'ENQEMV | 0 | | U-Boot / uImage. Das U-Boot Universal Boot Loader.[70] |
| 7B 5C 72 74 66 31 | {\rtf1 | 0 | rtf | Rich Text Format |
| 54 41 50 45 | TAPE | 0 | | Microsoft Tape Format |
| 47 | G | 0<br>0xBC<br>0x178<br>...<br>(every 188th byte) | ts<br>tsv<br>tsa<br>mpg<br>mpeg | MPEG Transport Stream (MPEG-2 Part 1)[71] |
| 00 00 01 BA | NULNULSOH° | 0 | m2p<br>vob<br>mpg<br>mpeg | MPEG Program Stream (MPEG-1 Part 1 (essentially identical) and MPEG-2 Part 1) |
| 00 00 01 B3 | NULNULSOH³ | 0 | mpg<br>mpeg | MPEG-1 video and MPEG-2 video (MPEG-1 Part 2 and MPEG-2 Part 2) |
| 66 74 79 70 69 73 6F 6D | ftypisom | 4 | mp4 | ISO Base Media file (MPEG-4) |
| 66 74 79 70 4D 53 4E 56 | ftypMSNV | 4 | mp4 | MPEG-4 video file |
| 78 01 | xSOH | 0 | zlib | No Compression (no preset dictionary) |
| 78 5E | x^ | | | Best speed (no preset dictionary) |
| 78 9C | xœ | | | Default Compression (no preset dictionary) |
| 78 DA | xÚ | | | Best Compression (no preset dictionary) |
| 78 20 | xSP | | | No Compression (with preset dictionary) |
| 78 7D | x} | | | Best speed (with preset dictionary) |
| 78 BB | x» | | | Default Compression (with preset dictionary) |
| 78 F9 | xù | | | Best Compression (with preset dictionary) |
| 62 76 78 32 | bvx2 | 0 | lzfse | LZFSE - Lempel-Ziv style data compression algorithm using Finite State Entropy coding. |

| Hex signature | ISO 8859-1 | Offset | Extension | Description |
|---|---|---|---|---|
| | | | | OSS by Apple.[72] |
| 4F 52 43 | ORC | 0 | orc | Apache ORC (Optimized Row Columnar) file format |
| 4F 62 6A 01 | Obj☺ | 0 | avro | Apache Avro binary file format |
| 53 45 51 36 | SEQ6 | 0 | rc | RCFile columnar file format |
| 3C 72 6F 62 6C 6F 78 21 | <roblox! | 0 | rbxl | Roblox place file [73] |
| 65 87 78 56 | e‡xV | 0 | p25<br>obt | PhotoCap Object Templates |
| 55 55 AA AA | UUªª | 0 | pcv | PhotoCap Vector |
| 78 56 34 | xV4 | 0 | pbt<br>pdt<br>pea<br>peb<br>pet<br>pgt<br>pict<br>pjt<br>pkt<br>pmt | PhotoCap Template |
| 50 41 52 31 | PAR1 | 0 | | Apache Parquet columnar file format |
| 45 4D 58 32 | EMX2 | 0 | ez2 | Emulator Emaxsynth samples |
| 45 4D 55 33 | EMU3 | 0 | ez3<br>iso | Emulator III synth samples |
| 1B 4C 75 61 | ←Lua | 0 | luac | Lua bytecode[74] |
| 62 6F 6F 6B 00 00 00 00<br>6D 61 72 6B 00 00 00 00 | book␀␀␀␀mark␀␀␀␀ | 0 | alias | macOS file Alias[75] (Symbolic link) |
| 62 6F 6F 6B | book | 0 | | macOS bookmark (https://mac-alias.readthedocs.io/en/latest/bookmark_fmt.html) format |
| 5B 5A 6F 6E 65 54 72 61<br>6E 73 66 65 72 5D | [ZoneTransfer] | 0 | Identifier | Microsoft Zone Identifier for URL Security Zones[76][77] |
| 52 65 63 65 69 76 65 64<br>3A | Received: | 0 | eml | Email Message var5 |

| Hex signature | ISO 8859-1 | Offset | Extension | Description |
|---|---|---|---|---|
| 20 02 01 62 A0 1E AB 07 02 00 00 00 | ␣␂␁b␣␞«␇␂␀␀␀ | 0 | tde | Tableau Datasource |
| 37 48 03 02 00 00 00 00 58 35 30 39 4B 45 59 | 7H␃␂␀␀␀␀X509KEY | 0 | kdb | KDB file |
| 85 ?? ?? 03 | …??␃ | 0 | pgp | PGP file [78] |
| 28 B5 2F FD | (µ/ý | 0 | zst | Zstandard compress[79] [80] |
| 52 53 56 4B 44 41 54 41 | RSVKDATA | 0 | rs | QuickZip rs compressed archive[81][82] |
| 3A 29 0A | :)␤ | 0 | sml | Smile file |
| 4A 6F 79 21 | Joy! | 0 | | Preferred Executable Format |
| 34 12 AA 55 | 4½ªU | 0 | vpk | VPK file, used to store game data for some Source Engine games |
| 2A 2A 41 43 45 2A 2A | **ACE** | 7 | ace | ACE (compressed file format) |
| 60 EA | `ê | 0 | arj | ARJ |
| 49 53 63 28 | ISc( | 0 | cab | InstallShield CAB Archive File |
| 4B 57 41 4A | KWAJ | 0 | ??_ | Windows 3.1x Compressed File |
| 53 5A 44 44 | SZDD | 0 | ??_ | Windows 9x Compressed File |
| 5A 4F 4F | ZOO | 0 | zoo | Zoo (file format) |
| 50 31 0A | P1␤ | 0 | pbm | Portable bitmap ASCII |
| 50 34 0A | P4␤ | 0 | pbm | Portable bitmap binary |
| 50 32 0A | P2␤ | 0 | pgm | Portable Gray Map ASCII |
| 50 35 0A | P5␤ | 0 | pgm | Portable Gray Map binary |
| 50 33 0A | P3␤ | 0 | ppm | Portable Pixmap ASCII |
| 50 36 0A | P6␤ | 0 | ppm | Portable Pixmap binary |
| D7 CD C6 9A | ×ÍÆš | 0 | wmf | Windows Metafile |
| 67 69 6D 70 20 78 63 66 | gimp xcf | 0 | xcf | XCF (file format) |
| 2F 2A 20 58 50 4D 20 2A 2F | /* XPM */ | 0 | xpm | X PixMap |

| Hex signature | ISO 8859-1 | Offset | Extension | Description |
|---|---|---|---|---|
| 41 46 46 | AFF | 0 | aff | Advanced Forensics Format |
| 45 56 46 32 | EVF2 | 0 | Ex01 | EnCase EWF version 2 format |
| 45 56 46 | EVF | 0 | e01 | EnCase EWF version 1 format |
| 51 46 49 | QFI | 0 | qcow | qcow file format |
| 52 49 46 46 ?? ?? ?? ?? 41 43 4F 4E | RIFF????ACON | 0 | ani | Animated cursor |
| 52 49 46 46 ?? ?? ?? ?? 43 44 44 41 | RIFF????CDDA | 0 | cda | .cda file |
| 52 49 46 46 ?? ?? ?? ?? 51 4C 43 4D | RIFF????QLCM | 0 | qcp | Qualcomm PureVoice file format |
| 52 49 46 58 ?? ?? ?? ?? 46 47 44 4D (big-endian) | RIFX????FGDM | 0 | dcr | Adobe Shockwave[83][84][85] |
| 58 46 49 52 ?? ?? ?? ?? 4D 44 47 46 (little-endian) | XFIR????MDGF | | | |
| 52 49 46 58 ?? ?? ?? ?? 4D 56 39 33 (big-endian) | RIFX????MV93 | 0 | dir<br>dxr<br>drx | Macromedia Director file format[86][84][85] |
| 58 46 49 52 ?? ?? ?? ?? 33 39 56 4D (little-endian) | XFIR????39VM | | | |
| 46 4C 56 | FLV | 0 | flv | Flash Video file |
| 3C 3C 3C 20 4F 72 61 63 6C 65 20 56 4D 20 56 69 72 74 75 61 6C 42 6F 78 20 44 69 73 6B 20 49 6D 61 67 65 20 3E 3E 3E | <<< Oracle VM VirtualBox Disk Image >>> | 0 | vdi | VirtualBox Virtual Hard Disk file format |
| 63 6F 6E 65 63 74 69 78 | conectix | 0 | vhd | Windows Virtual PC Virtual Hard Disk file format[87] |

| Hex signature | ISO 8859-1 | Offset | Extension | Description |
|---|---|---|---|---|
| 76 68 64 78 66 69 6C 65 | vhdxfile | 0 | vhdx | Windows Virtual PC Windows 8 Virtual Hard Disk file format |
| 49 73 5A 21 | IsZ! | 0 | isz | Compressed ISO image |
| 44 41 41 | DAA | 0 | daa | Direct Access Archive PowerISO |
| 4C 66 4C 65 | LfLe | 0 | evt | Windows Event Viewer file format |
| 45 6C 66 46 69 6C 65 | ElfFile | 0 | evtx | Windows Event Viewer XML file format |
| 73 64 62 66 | sdbf | 8 | sdb | Windows customized database |
| 50 4D 43 43 | PMCC | 0 | grp | Windows 3.x Program Manager Program Group file format |
| 4B 43 4D 53 | KCMS | 0 | icm | ICC profile |
| 72 65 67 66 | regf | 0 | dat<br>hiv | Windows Registry file |
| 21 42 44 4E | !BDN | 0 | pst | Microsoft Outlook Personal Storage Table file |
| 44 52 41 43 4F | DRACO | 0 | drc | 3D model compressed with Google Draco[88] |
| 47 52 49 42 | GRIB | 0 | grib<br>grib2 | Gridded data (commonly weather observations or forecasts) in the WMO GRIB or GRIB2 format[89] |
| 42 4C 45 4E 44 45 52 | BLENDER | 0 | blend | Blender File Format[90] |
| 00 00 00 0C 4A 58 4C 20 0D 0A 87 0A | ␀␀␀␌JXL␣␍␊‡␊ | 0 | jxl | Image encoded in the JPEG XL format[91] |
| FF 0A | ÿ␊ | | | |
| 00 01 00 00 00 | ␀␁␀␀␀ | 0 | ttf<br>tte<br>dfont | TrueType font |
| 4F 54 54 4F | OTTO | 0 | otf | OpenType font[92] |
| 23 25 4D 6F 64 75 6C 65 | #%Module | 0 | | Modulefile for Environment Modules[93] |
| 4D 53 57 49 4D 00 00 00 D0 00 00 00 00 | MSWIM␀␀␀Ð␀␀␀␀ | 0 | wim<br>swm<br>esd | Windows Imaging Format file |

| Hex signature | ISO 8859-1 | Offset | Extension | Description |
|---|---|---|---|---|
| 21 2D 31 53 4C 4F 42 1F | !-1SLOB$_{US}$ | 0 | slob | Slob (sorted list of Object storages) is a read-only, compressed data store with dictionary-like interface[94] |
| AC ED | ¬í | 0 | | Serialized Java Data[95] |
| 43 72 65 61 74 69 76 65 20 56 6F 69 63 65 20 46 69 6C 65 1A 1A 00 | Creative Voice File | 0 | voc | Creative Voice file |
| 2E 73 6E 64 | .snd | 0 | au<br>snd | Au audio file format |
| DB 0A CE 00 | | 0 | | OpenGL Iris Perfomer .PFB (Performer Fast Binary)[96] |
| 48 5a 4c 52 00 00 00 18 | HZLR | 0 | hazelrules | Noodlesoft Hazel [97] |
| 46 4C 68 64 | FLhd | 0 | flp | FL Studio Project File |
| 31 30 4C 46 | 10LF | 0 | flm | FL Studio Mobile Project File |
| 52 4b 4d 43 32 31 30 | RKMC210 | 0 | | Vormetric Encryption DPM Version 2.1 Header[98] |
| 00 01 00 00 4D 53 49 53 41 4D 20 44 61 74 61 62 61 73 65 | $_{NULNULNULNUL}$MSISAM Database | 0 | mny | Microsoft Money file |
| 00 01 00 00 53 74 61 6E 64 61 72 64 20 41 43 45 20 44 42 | $_{NULSOHNULNUL}$Standard ACE DB | 0 | accdb | Microsoft Access 2007 Database |
| 00 01 00 00 53 74 61 6E 64 61 72 64 20 4A 65 74 20 44 42 | $_{NULSOHNULNUL}$Standard Jet DB | 0 | mdb | Microsoft Access Database |
| 01 FF 02 04 03 02 | $_{SOH}$ÿ$_{STX}$$_{EOT}$$_{ETX}$$_{STX}$ | 0 | drw | Micrografx vector graphic file |
| 02 64 73 73 | $_{STX}$dss | 0 | dss | Digital Speech Standard (Olympus, Grundig, & Phillips) v2 |
| 03 64 73 73 | $_{ETX}$dss | 0 | dss | Digital Speech Standard (Olympus, Grundig, & Phillips) v3 |
| 03 00 00 00 41 50 50 52 | $_{ETX}$$_{NUL}$$_{NUL}$$_{NUL}$APPR | 0 | adx | Approach index file |

| Hex signature | ISO 8859-1 | Offset | Extension | Description |
|---|---|---|---|---|
| 06 06 ED F5 D8 1D 46 E5 BD 31 EF E7 FE 74 B7 1D | ♠♠íõØ♦Få½1ïçþt·♦ | 0 | indd | Adobe InDesign document |
| 06 0E 2B 34 02 05 01 01 0D 01 02 01 01 02 | ♠♦+4♦♦♦♦♦♦♦♦♦♦♦♦ | 0-65535 (run-in) | mxf | Material Exchange Format file |
| 07 53 4B 46 | •SKF | 0 | skf | SkinCrafter skin file |
| 07 64 74 32 64 64 74 64 | •dt2ddtd | 0 | dtd | DesignTools 2D Design file |
| 0A 16 6F 72 67 2E 62 69 74 63 6F 69 6E 2E 70 72 | ♪♦org.bitcoin.pr | 0 | wallet | MultiBit Bitcoin wallet file |
| 0D 44 4F 43 | ♪DOC | 0 | doc | DeskMate Document file |
| 0E 4E 65 72 6F 49 53 4F | ♫NeroISO | 0 | nri | Nero CD Compilation |
| 0E 57 4B 53 | ♫WKS | 0 | wks | DeskMate Worksheet |
| 0F 53 49 42 45 4C 49 55 53 | ☼SIBELIUS | 0 | sib | Sibelius Music - Score file |
| 23 20 4D 69 63 72 6F 73 6F 66 74 20 44 65 76 65 6C 6F 70 65 72 20 53 74 75 64 69 6F | # Microsoft Developer Studio | 0 | dsp | Microsoft Developer Studio project file |
| 23 21 41 4D 52 | #!AMR | 0 | amr | Adaptive Multi-Rate ACELP (Algebraic Code Excited Linear Prediction) Codec, commonly audio format with GSM cell phones. |
| 23 21 53 49 4C 4B 0A | #!SILK♪ | 0 | sil | Audio compression format developed by Skype |
| 23 3F 52 41 44 49 41 4E 43 45 0A | #?RADIANCE♪ | 0 | hdr | Radiance High Dynamic Range image file |
| 23 40 7E 5E | #@~^ | 0 | vbe | VBScript Encoded script |
| 0D F0 1D C0 | ♪ð▯À | 0 | cdb | MikroTik WinBox Connection Database (Address Book) |
| 23 45 58 54 4D 33 55 | #EXTM3U | 0 | m3u<br>m3u8 | Multimedia playlist |

| Hex signature | ISO 8859-1 | Offset | Extension | Description |
|---|---|---|---|---|
| 6D 64 66 00 | mdf␀ | 0 | m | M2 Archive, used by game developer M2 |
| 4B 50 4B 41 | KPKA | 0 | pak | Capcom RE Engine game data archives |
| 41 52 43 | ARC | 0 | arc | Capcom MT Framework game data archives |
| 41 72 43 | ArC | 0 | arc<br>cdx | FreeArc file |
| D0 4F 50 53 | ÐOPS | 0 | pl | Interleaf PrinterLeaf / WorldView document format (now Broadvision QuickSilver) |
| 6E 2B 31 00 | n+1 | 344 | nii | Single file NIfTI format, used extensively in biomedical imaging. |
| 6E 69 31 00 | ni1 | 344 | hdr | Header file of a .hdr/.img pair in NIfTI format, used extensively in biomedical imaging. |
| 52 41 46 36 34 | RAF64 | 0 | | Report Builder file from Digital Metaphors (https://www.digital-metaphors.com/) |
| 56 49 53 33 | VIS3 | 0 | | Resource file Visionaire 3.x Engine (https://www.visionaire-studio.net/) |
| 4D 53 48 7C<br>42 53 48 7C | MSH\|<br>BSH\| | 0 | hl7 | Health Level Seven (HL7) Standard for electronic data exchange [1] (https://www.hl7.eu/HL7v2x/v25/std25/hl7.html) |
| 70 77 72 64 61 74 61 | pwrdata | 0 | pwrdata | SAP Power Monitor (https://github.com/SAP/power-monitoring-tool-for-macos) (version 1.1.0 and higher) data file |
| 1a 08 | .. | 0 | arc | ARC archive file |
| 2d 2d 2d 2d 2d 42 45 47 49 4e 20 50 47 50 20 50 55 42 4c 49 43 20 4b 45 49 20 42 4c 4f 43 4b 2d 2d 2d 2d 2d | -----BEGIN PGP PUBLIC KEY BLOCK----- | 0 | asc | Armored PGP public key |
| 3a 42 61 73 65 20 | :Base | 0 | cnt | Windows 3.x - Windows 95 Help Contents |

| Hex signature | ISO 8859-1 | Offset | Extension | Description |
|---|---|---|---|---|
| 52 49 46 46 ?? ?? ?? ?? 56 44 52 4d | RIFF????VDRM | 0 | vdr | VirtualDub |
| 52 59 46 46 ?? ?? ?? ?? 54 52 49 44 | RIFF????TRID | 0 | trd | TrID |
| 52 49 46 46 ?? ?? ?? ?? 73 68 77 34 | RIFF????shw4 | 0 | shw | Corel SHOW! 4.0 |
| 52 49 46 46 ?? ?? ?? ?? 73 68 77 35 | RIFF????shw5 | 0 | shw | Corel SHOW! 5.0 |
| 52 49 46 46 ?? ?? ?? ?? 73 68 72 35 | RIFF????shr5 | 0 | shr | Corel SHOW! 5.0 player |
| 52 49 46 46 ?? ?? ?? ?? 73 68 62 35 | RIFF????shb5 | 0 | shb | Corel SHOW! 5.0 background |
| 52 49 46 46 ?? ?? ?? ?? 52 4d 4d 50 | RIFF????RMMP | 0 | mmm | MacroMind Multimedia Movie or Microsoft Multimedia Movie |
| 41 53 54 4d 2d 45 35 37 | ASTM-E57 | 0 | e57 | ASTM E57 3D file format |
| aa aa aa aa | ªªªª | 0 | sys | Crowdstrike Channel File |
| 8C 0A 00 | Œ.. | 0 | ucas | Unreal Engine Compressed Asset Storage file |
| 2D 3D 3D 2D 2D 3D 3D 2D 2D 3D 3D 2D 2D 3D 3D 2D | -==--==--==--==- | 0 | utoc | Unreal Engine Table of Contents file |
| 43 36 34 46 69 6C 65 00 | C64File | 0 | P00 (P01, P02,...) | Commodore 64 binary file (source: *.P00 format for Power64 emulator (https://www.salto.at/Power64/Documentation/Power64-Leggimi/AE-Formati_File.html#Section%20E.1)) |
| 44 55 43 4B | DUCK | 8 | | DuckDB database file (source: Source code (https://github.com/duckdb/duckdb/blob/main/sr |

| Hex signature | ISO 8859-1 | Offset | Extension | Description |
|---|---|---|---|---|
| | | | | c/include/duckdb/storage/storage_info.hpp#L69)) |
| 4C 50 4B 53 48 48 52 48 | LPKSHHRH | 0 | journal | journald log file[99] |
| E0 01 00 EA | ê␑␀ê | 0 | | Ion (serialization format) binary encoding[100] |
| 42 53 44 49 46 46 34 30 | BSDIFF40 | 0 | | bsdiff binary patch |
| 7C 61 8A B2 | \|aŠ² | 0 | hpapp<br>hpprgm<br>hpsettings | HP Prime graphing calculator file |
| 43 5A 50 58 | CZPX | 0 | czx | Czip X (https://czip.it) encrypted archive (v4.x) |
| 43 50 43 48 | CPCH | 0 | pch | Clang pre-compiled header |

## See also

- List of filename extensions - alternative for file type identification and parsing
- List of file formats
- Magic number (programming)
- Substitute character (for the 1Ah (^Z) "end-of-file" marker used in many signatures)
- file (command)

## References


1. "execve(2): execute program - Linux man page" (https://linux.die.net/man/2/execve). *linux.die.net*. Retrieved 2022-07-12.
2. "Libpcap File Format" (https://wiki.wireshark.org/Development/LibpcapFileFormat#Global_Header). Retrieved 2018-06-19.
3. "PCAP Next Generation Dump File Format" (https://www.winpcap.org/ntar/draft/PCAP-DumpFileFormat.html#sectionshb). Retrieved 2018-06-19.
4. "A. Format of the RPM file" (ftp://ftp.tuwien.ac.at/.vhost/www.openpkg.org/doc/book/maximum-rpm.html/node26.html). *FTP server* (FTP). Retrieved 2017-02-15. (To view documents see Help:FTP)
5. "Database File Format" (https://www.sqlite.org/fileformat.html). Retrieved 2018-11-16.
6. "GitHub - NiLuJe/KindleTool: Tool for creating/extracting Kindle updates and more" (https://github.com/NiLuJe/KindleTool). *GitHub*. Retrieved 2017-02-15.
7. "IWAD" (https://zdoom.org/wiki/IWAD). Retrieved 2022-07-05.
8. Icons (http://msdn.microsoft.com/en-us/library/ms997538.aspx) (at MSDN)

9. GRAPHICS INTERCHANGE FORMAT(sm) Version 89a (http://www.w3.org/Graphics/GIF/spec-gif89a.txt)
10. "TIFF, Revision 6.0" (https://www.loc.gov/preservation/digital/formats/fdd/fdd000022.shtml). *Sustainability of Digital Formats: Planning for Library of Congress Collections*. The Library of Congress. 2022-04-06. Retrieved 2022-07-12.
11. "BigTIFF" (https://www.loc.gov/preservation/digital/formats/fdd/fdd000328.shtml). *loc.gov*. 2024-04-15. Retrieved 2024-09-20.
12. Computer Knowledge. "File Extension .CR2 Details" (http://filext.com/file-extension/CR2). *filext.com*.
13. "Inside the Canon RAW format version 2, understanding .CR2 file format and files produced by Canon EOS Digital Camera" (http://lclevy.free.fr/cr2/#key_info). *free.fr*.
14. "Rob Northen compression" (https://segaretro.org/Rob_Northen_compression). *Sega Retro*. 11 August 2020. Retrieved 18 January 2024.
15. "domsson/nuru" (https://github.com/domsson/nuru). *github.com*. 9 September 2023.
16. Bellard, Fabrice (2015). "BPG Image format" (http://bellard.org/bpg/).
17. "Overview of JPEG 1" (https://jpeg.org/jpeg/).
18. "Overview of JPEG 2000" (https://jpeg.org/jpeg2000/).
19. "qoi-specification" (https://qoiformat.org/qoi-specification.pdf) (PDF).
20. "Lzip Compressed Format and the 'application/lzip' Media Type" (https://datatracker.ietf.org/doc/draft-diaz-lzip/). *Ietf Datatracker*. section 2.
21. "Format of cpio archives" (https://www.mkssoftware.com/docs/man4/cpio.4.asp). Retrieved 2022-07-05.
22. "Format of SmartSniff packets file" (https://www.nirsoft.net/utils/smsniff.html). Retrieved 2024-08-05.
23. *TechNote.txt: RAR version 4.00 - Technical information* (Technical report). 2010-12-01. "The marker block is actually considered as a fixed byte sequence: 0x52 0x61 0x72 0x21 0x1a 0x07 0x00"
24. "RAR 5.0 archive format" (https://www.rarlab.com/technote.htm). Retrieved 2023-10-19.
25. Tool Interface Standard (TIS) *Executable and Linking Format (ELF) Specification (http://refspecs.linuxbase.org/elf/elf.pdf) Version 1.2* (May 1995)
26. *PNG (Portable Network Graphics) Specification Version 1.0* (https://www.rfc-editor.org/rfc/rfc2083). IETF. doi:10.17487/RFC2083 (https://doi.org/10.17487%2FRFC2083). RFC 2083 (https://datatracker.ietf.org/doc/html/rfc2083).
27. Elliott, John C.; Lopushinsky, Jim (2002) [1998-04-11]. "CP/M 3.0 COM file header" (http://www.seasip.info/Cpm/rsxrec.html). *Seasip.info*. Archived (https://web.archive.org/web/20160830194033/http://www.seasip.info/Cpm/rsxrec.html) from the original on 2016-08-30. Retrieved 2016-08-29.
28. "Faq - Utf-8, Utf-16, Utf-32 & Bom" (https://unicode.org/faq/utf_bom.html#BOM).
29. "How to : Load XML from File with Encoding Detection" (https://odieweblog.wordpress.com/2016/04/10/how-to-load-xml-from-file-with-encoding-detection). 10 April 2016.
30. "SDL Documentation" (https://docs.sdl.com/791187/581899/sdl-contenta-5-7/representations-of-boms-by-encoding).

31. Honerman, Tom (January 2, 2021). "Clarify guidance for use of a BOM as a UTF-8 encoding signature" (https://unicode.org/L2/L2021/21038-bom-guidance.pdf) (PDF).
32. "JKS" (https://hg.openjdk.java.net/jdk10/jdk10/jdk/file/777356696811/src/java.base/share/classes/sun/security/provider/JavaKeyStore.java). *Javakey Store*.
33. "Mac Developer Library" (https://developer.apple.com/library/mac/#documentation/DeveloperTools/Conceptual/MachORuntime/Reference/reference.html). *apple.com*.
34. "Encapsulated PostScript (EPS) File Format, Version 3.x" (https://www.loc.gov/preservation/digital/formats/fdd/fdd000246.shtml). *Library of Congress*. 10 May 2022. Retrieved 2022-07-05.
35. "Encapsulated PostScript (EPS) File Format, Version 3.x" (https://www.loc.gov/preservation/digital/formats/fdd/fdd000246.shtml). *Library of Congress*. 10 May 2022. Retrieved 2022-07-05.
36. "File command PDF Magic format" (https://github.com/file/file/blob/master/magic/Magdir/pdf). *GitHub*. Retrieved 2018-11-06.
37. "ASF (Advanced Systems Format)" (http://www.digitalpreservation.gov/formats/fdd/fdd000067.shtml). 6 April 2007.
38. "WAVE Audio File Format" (https://www.loc.gov/preservation/digital/formats/fdd/fdd000001.shtml). *Sustainability of Digital Formats: Planning for Library of Congress Collections*. The Library of Congress. 2022-04-19. Retrieved 2022-07-12.
39. "AVI (Audio Video Interleaved) File Format" (https://www.loc.gov/preservation/digital/formats/fdd/fdd000059.shtml). *Sustainability of Digital Formats: Planning for Library of Congress Collections*. The Library of Congress. 2016-03-09. Retrieved 2022-07-12.
40. Gary C. Kessler (21 October 2012). "File Signatures Table" (http://www.garykessler.net/library/file_sigs.html). Retrieved 28 December 2012.
41. "Flexible Image Transport System (FITS), Version 3.0 - File type signifiers" (https://www.loc.gov/preservation/digital/formats/fdd/fdd000317.shtml#sign). *Library of Congress*. 2012-09-26. Retrieved 16 June 2021.
42. Josh Coalson. "FLAC— format" (https://xiph.org/flac/format.html#stream). Retrieved 16 June 2021. " "fLaC", the FLAC stream marker in ASCII, meaning byte 0 of the stream is 0x66, followed by 0x4C 0x61 0x43"
43. "File Signature Database: mid File Signatures" (https://filesignatures.net/index.php?search=mid&mode=EXT). *filesignatures.net*.
44. "Developing a tool to recognise MS Office file types ( .doc, .xls, .mdb, .ppt )" (https://web.archive.org/web/20140809205308/http://social.msdn.microsoft.com/Forums/en-US/343d09e3-5fdf-4b4a-9fa6-8ccb37a35930/developing-a-tool-to-recognise-ms-office-file-types-doc-xls-mdb-ppt-?forum=os_binaryfile). *social.msdn.microsoft.com*. Archived from the original (http://social.msdn.microsoft.com/Forums/en-US/343d09e3-5fdf-4b4a-9fa6-8ccb37a35930/developing-a-tool-to-recognise-ms-office-file-types-doc-xls-mdb-ppt-?forum=os_binaryfile) on 2014-08-09. Retrieved 2014-07-28.
45. Gary C. Kessler, Ph.D., CCE, CISSP. "File Signatures" (https://web.archive.org/web/20221112073316/https://www.garykessler.net/library/file_sigs.html). *www.garykessler.net*. Archived from the original (https://www.garykessler.net/library/file_sigs.html) on 2022-11-12. Retrieved 2025-12-05. "Note the similarity between D0 CF 11 E0 and the word "DOCFILE"!"

46. "What Files Make Up a Virtual Machine?" (https://web.archive.org/web/20161227211713/https://www.vmware.com/support/ws55/doc/ws_learning_files_in_a_vm.html/). *VMware*. Archived from the original (https://www.vmware.com/support/ws55/doc/ws_learning_files_in_a_vm.html) on 2016-12-27. Retrieved 2014-02-22.
47. "VMware Virtual Disks Virtual Disk Format 1.1" (https://web.archive.org/web/20111015053621/http://www.vmware.com/app/vmdk/?src=vmdk). *VMware*. Archived from the original (http://www.vmware.com/app/vmdk/?src=vmdk) on 2011-10-15. Retrieved 2014-02-22.
48. "CRX Package Format" (https://developer.chrome.com/extensions/crx.html). *chrome.com*.
49. "CRX Package Format" (https://web.archive.org/web/20131216082442/http://developer.chrome.com/apps/crx.html). *chrome.com*. Archived from the original (https://developer.chrome.com/apps/crx.html) on 2013-12-16. Retrieved 2013-12-16.
50. "[Pythonmac-SIG] Discovering file type" (https://mail.python.org/pipermail/pythonmac-sig/2005-February/013028.html). *python.org*. 10 February 2005.
51. Kehl, Ken. "Re: What is the suffix for Freehand files?" (https://groups.google.com/g/macromedia.freehand/c/g6bGwP-YYks/m/4m2Xfk--7VkJ). *Google Groups*.
52. "xar - xarformat.wiki" (https://code.google.com/p/xar/wiki/xarformat). *code.google.com*.
53. "Easily Restore Your Computer With File and Settings Transfer Wizard XP (Part 1)" (https://www.howtogeek.com/79820/easily-restore-your-computer-with-file-and-settings-transfer-wizard-xp-part-1/). *howtogeek.com*. 27 August 2007.
54. "User State Migration Tool 3.0" (https://technet.microsoft.com/en-us/library/cc722032%28v=ws.10%29.aspx). *microsoft.com*. Microsoft.
55. "User State Migration Tool 4.0 User's Guide" (https://technet.microsoft.com/en-us/library/dd560801.aspx). *microsoft.com*. Microsoft. 29 June 2010.
56. "NESRomTool::NES ROM Quickstart" (http://sadistech.com/nesromtool/romdoc.html). *sadistech.com*.
57. "GNU tar 1.28: Basic Tar Format" (https://www.gnu.org/software/tar/manual/html_node/Standard.html). *gnu.org*.
58. "UVOX Universal Voxel Translator - Man Page" (https://web.archive.org/web/20160508202038/http://tox.land/uvox/man.html). *tox.land*. Archived from the original (http://tox.land/uvox/man.html) on 2016-05-08. Retrieved 2015-06-29.
59. "RAW Format v2.0 - Google Sheets" (https://docs.google.com/spreadsheet/ccc?key=0AgQ2MOkAZTFHdHJraTVTOEpmNEIwTVlKd0dHVi1ULUE#gid=0). *google.com*.
60. "Using Binary Delta Compression (BDC) Technology to Update Windows Operating Systems" (http://www.microsoft.com/en-us/download/details.aspx?id=1562). Microsoft. 10 June 2005. Retrieved 15 March 2015.
61. Deutsch, L. Peter (May 1996). "Member header and trailer" (https://datatracker.ietf.org/doc/html/rfc1952#page-6). *GZIP file format specification version 4.3* (https://www.rfc-editor.org/rfc/rfc1952). p. 6. doi:10.17487/RFC1952 (https://doi.org/10.17487%2FRFC1952). RFC 1952 (https://datatracker.ietf.org/doc/html/rfc1952). Retrieved 2018-10-29.
62. Collet, Yann (19 November 2021). "LZ4 Frame format" (https://github.com/lz4/lz4/blob/dev/doc/lz4_Frame_format.md). *github.com*.
63. Collet, Yann (19 November 2021). "LZ4 Block Format" (https://github.com/lz4/lz4/blob/dev/doc/lz4_Block_format.md). *github.com*.

64. "WOFF 1" (https://www.w3.org/TR/2012/REC-WOFF-20121213/). *w3.org*. 13 December 2012. Retrieved 18 January 2024.
65. "WOFF 2" (https://www.w3.org/TR/WOFF2/). *w3.org*. 13 December 2012. Retrieved 18 January 2024.
66. "Extensible Markup Language (XML) 1.0 (Fifth Edition)" (https://www.w3.org/TR/xml/#sec-guessing-no-ext-info).
67. "WebAssembly/design" (https://github.com/WebAssembly/design/blob/master/BinaryEncoding.md#high-level-structure). *GitHub*. Retrieved 2016-11-01.
68. "Lepton image compression: saving 22% losslessly from images at 15MB/s" (https://blogs.dropbox.com/tech/2016/07/lepton-image-compression-saving-22-losslessly-from-images-at-15mbs/). *Dropbox*.
69. "Journal (jdb2)" (https://web.archive.org/web/20250811171405/https://www.kernel.org/doc/html/latest/filesystems/ext4/journal.html#block-header). *Linux kernel*. Archived from the original (https://www.kernel.org/doc/html/latest/filesystems/ext4/journal.html#block-header) on 2025-08-11. Retrieved 2025-10-21.
70. "How to Extract an uImage" (https://web.archive.org/web/20170225051131/http://buffalo.nas-central.org/wiki/How_to_Extract_an_uImage). Archived from the original (http://buffalo.nas-central.org/wiki/How_to_Extract_an_uImage) on 2017-02-25. Retrieved 2017-02-24.
71. "An introduction to MPEG-TS" (https://tsduck.io/download/docs/mpegts-introduction.pdf) (PDF).
72. "lzfse/lzfse" (https://github.com/lzfse/lzfse). *GitHub*. Retrieved 2018-10-16.
73. "ROBLOX File Format" (https://www.classy-studios.com/Downloads/RobloxFileSpec.pdf) (PDF). *classy-studios.com*. Retrieved 5 September 2023.
74. Laurie, Dirk (2013). "Lua 5.2 Bytecode and Virtual Machine" (http://files.catwell.info/misc/mirror/lua-5.2-bytecode-vm-dirk-laurie/lua52vm.html).
75. "Mac OS X - ForensicsWiki" (https://web.archive.org/web/20180922173418/https://www.forensicswiki.org/wiki/Mac_OS_X#Burn_Folder). *forensicswiki.org*. Archived from the original (https://www.forensicswiki.org/wiki/Mac_OS_X#Burn_Folder) on 2018-09-22. Retrieved 2018-09-22.
76. "About URL Security Zones (Windows)" (https://technet.microsoft.com/en-us/windows/ms537183(v=vs.60)). *learn.microsoft.com*. 15 August 2017. Retrieved 18 January 2024.
77. "libyal/libfsntfs" (https://github.com/libyal/libfsntfs/blob/master/documentation/New%20Technologies%20File%20System%20(NTFS).asciidoc). *GitHub*. Retrieved 2018-09-22.
78. "gnupg - Is it a coincidence that the first 4 bytes of a PGP/GPG file are ellipsis, smile, female sign and a heart? - Information Security Stack Exchange)" (https://security.stackexchange.com/a/144555/12109). *security.stackexchange.com*. Retrieved 2020-06-05.
79. Collet, Yann; Kucherawy, Murray (October 2018). "Zstandard Frames" (https://datatracker.ietf.org/doc/html/rfc8478#section-3.1.1). *Zstandard Compression and the application/zstd Media Type* (https://www.rfc-editor.org/rfc/rfc8478). sec. 3.1.1. doi:10.17487/RFC8478 (https://doi.org/10.17487%2FRFC8478). RFC 8478 (https://datatracker.ietf.org/doc/html/rfc8478). Retrieved 2018-10-29.
80. "Magic numbering scheme (zstd project issue tracker)" (https://github.com/facebook/zstd/issues/768). *GitHub*. Retrieved 2018-10-29.

81. "RS File Extension" (https://filext.com/file-extension/RS). *filext.com*. Retrieved 2020-09-18.
82. "Quickzip.org (WebArchive)" (https://web.archive.org/web/20050403233040/http://www.quickzip.org/help/). Archived from the original (http://www.quickzip.org/help/) on 2005-04-03. Retrieved 2020-09-18.
83. "DCR format - reverseShockwave" (https://sites.google.com/site/reverseshockwave/home/dcr-format).
84. "Shockwave (Director)" (https://web.archive.org/web/20200211184445/http://fileformats.archiveteam.org/wiki/Shockwave_(Director)). *fileformats.archiveteam.org*. Archived from the original (http://fileformats.archiveteam.org/wiki/Shockwave_(Director)) on 11 February 2020. Retrieved 15 January 2022.
85. "Projector EXE to Shockwave DCR" (https://docs.google.com/presentation/d/1pc-f2b4uB0qTFsvQ9XayQt3NYd6ARClPSS8ikOLtzbU/htmlpresent).
86. "Dir format - reverseShockwave" (https://sites.google.com/site/reverseshockwave/home/dir-format).
87. "Virtual Hard Disk Image Format Specification" (https://download.microsoft.com/download/f/f/e/ffef50a5-07dd-4cf8-aaa3-442c0673a029/Virtual%20Hard%20Disk%20Format%20Spec_10_18_06.doc).
88. "Draco Bitstream Specification" (https://google.github.io/draco/spec/). *google.github.io*. Archived (https://web.archive.org/web/20210118162812/https://google.github.io/draco/spec/) from the original on 2021-01-18. Retrieved 2021-01-18.
89. "A GUIDE TO THE CODE FORM FM 92-IX Ext. GRIB" (https://web.archive.org/web/20210502163758/https://www.wmo.int/pages/prog/www/WDM/Guides/Guide-binary-2.html). Archived from the original (https://www.wmo.int/pages/prog/www/WDM/Guides/Guide-binary-2.html) on 2021-05-02. Retrieved 2021-05-02.
90. "The mystery of the blend" (https://web.archive.org/web/20150418144220/http://www.atmind.nl/blender/mystery_ot_blend.html). 2015-04-18. Archived from the original (https://www.atmind.nl/blender/mystery_ot_blend.html) on 2015-04-18. Retrieved 2021-05-07.
91. "GitHub - JPEG/JPEG XL Reference Software" (https://github.com/libjxl/libjxl/blob/c98f133f3f5e456caaa2ba00bc920e923b713abc/lib/jxl/decode.cc#L107-L138). *github.com*. Retrieved 2021-08-11.
92. "The OpenType Font File" (https://docs.microsoft.com/en-us/typography/opentype/spec/otff). *docs.microsoft.com*. 9 December 2021. Retrieved 2022-02-07.
93. "Environment Modules Documentation: modulefile" (https://modules.readthedocs.io/en/latest/modulefile.html). Retrieved 2021-08-19.
94. *GitHub - itkach/slob: Data store for Aard 2 (https://github.com/itkach/slob)*
95. "Java Object Serialization Specification: 6 - Object Serialization Stream Protocol" (https://docs.oracle.com/javase/7/docs/platform/serialization/spec/protocol.html).
96. "J3k0/OpenPFB" (https://github.com/j3k0/OpenPFB/blob/40d3bb13f672f50beca921aea581684d218fd989/OpenPfb.h#L370). *GitHub*. 25 June 2022.
97. "Noodlesoft – Noodlesoft – Simply Useful Software" (https://www.noodlesoft.com/). *noodlesoft.com*. Retrieved 2022-07-24.
98. IBM. "Thales Vormetric Data Security Platform" (https://www.ibm.com/support/pages/system/files/inline-files/$FILE/VDS_6.2.0_VAE_Install_API_Guide.pdf) (PDF). *IBM*.

99. "Journal File Format" (https://systemd.io/JOURNAL_FILE_FORMAT/). *systemd.io*. Retrieved 2025-08-03.
100. "Ion Binary Encoding" (https://amazon-ion.github.io/ion-docs/docs/binary.html#value-streams). Retrieved 2025-08-07.

## External links

- Gary Kessler's list of file signatures (http://www.garykessler.net/library/file_sigs.html)
- Online File Signature Database for Forensic Practitioners, a private compilation free to Law Enforcement (http://www.filesig.co.uk/ofsdb.html)
- Man page for compress, uncompress, and zcat on SCO Open Server (https://web.archive.org/web/20100502014229/http://docsrv.sco.com:507/en/man/html.C/compress.C.html)
- Public Database of File Signatures (https://web.archive.org/web/20120621000045/http://www.filesignatures.net/)
- Complete list of magic numbers with sample files (http://asecuritysite.com/forensics/magic)
- the original libmagic data files with thousands of entries (https://github.com/file/file/tree/master/magic/Magdir) as used by file (command)

Retrieved from "https://en.wikipedia.org/w/index.php?title=List_of_file_signatures&oldid=1340251992"